

Search ID	Database	Date	Exact Search String	Filters Applied	Total Results	Results Reviewed	Sources Retained	Notes
A1	Google Scholar	16/06/2026	("window system" OR "residential window") (repair OR maintenance OR refurbishment OR replacement) ("service life" OR durability OR circularity)	2015-2026; English-language sources prioritised	Approximately 1,030	20	15	The first 20 results were reviewed in relevance order. Five sources were marked Keep, ten were marked Maybe, four were excluded, and one duplicate was identified. Sources were retained when they addressed window repair, maintenance, refurbishment, component replacement, service life, or transferable lifecycle evidence. Sources focused only on operational energy or unrelated window technologies were excluded.
B1	Google Scholar	18/06/2026	("building component" OR "window component" OR "window system") ("design for disassembly" OR modularity OR replaceability OR "component accessibility") (reuse OR repair OR "life extension" OR "product life extension")	2015-2026; English-language sources prioritised	Approximately 1,430	20	20	The first 20 results were reviewed in relevance order. Thirteen sources were marked Keep and seven were marked Maybe. No sources were excluded at this stage because all results provided relevant evidence on design for disassembly, component accessibility, reuse, lifecycle assessment, or the wider implementation conditions needed to support circular building components. The Maybe sources will be reassessed during full-text screening because some are broader than the residential window-system focus or overlap with stronger publications.
C1	Google Scholar	18/06/2026	("rebound effect" OR "unintended consequence" OR "burden shifting" OR "lifecycle trade-off") (repair OR circularity OR "product life extension" OR "product lifetime extension")	2015-2026; English-language sources prioritised	Approximately 16,500	10	10	The first 10 results were reviewed in relevance order. Five sources were marked Keep and five were marked Maybe. Sources were retained when they provided relevant evidence on circular-economy rebound effects, product-life extension, behavioural responses, strategy interactions, mitigation, lifecycle trade-offs, or burden shifting. The Maybe sources will be reassessed during full-text screening because some are not directly related to building components, overlap with stronger sources, or are not currently accessible in full text.
D1	Google Scholar	18/06/2026	("life cycle assessment" OR "lifecycle comparison") window (repair OR replacement OR refurbishment)	2015-2026; English-language sources prioritised	Approximately 17,900	10	6	The first 10 results were reviewed in relevance order. Four sources were marked Keep, two were marked Maybe, three were excluded, and one duplicate was identified. Sources were retained when they directly compared window maintenance, refurbishment, upgrading, or replacement, or when they provided a transferable lifecycle method for assessing retained and newly added components. Whole-building studies without sufficiently specific window evidence were excluded.